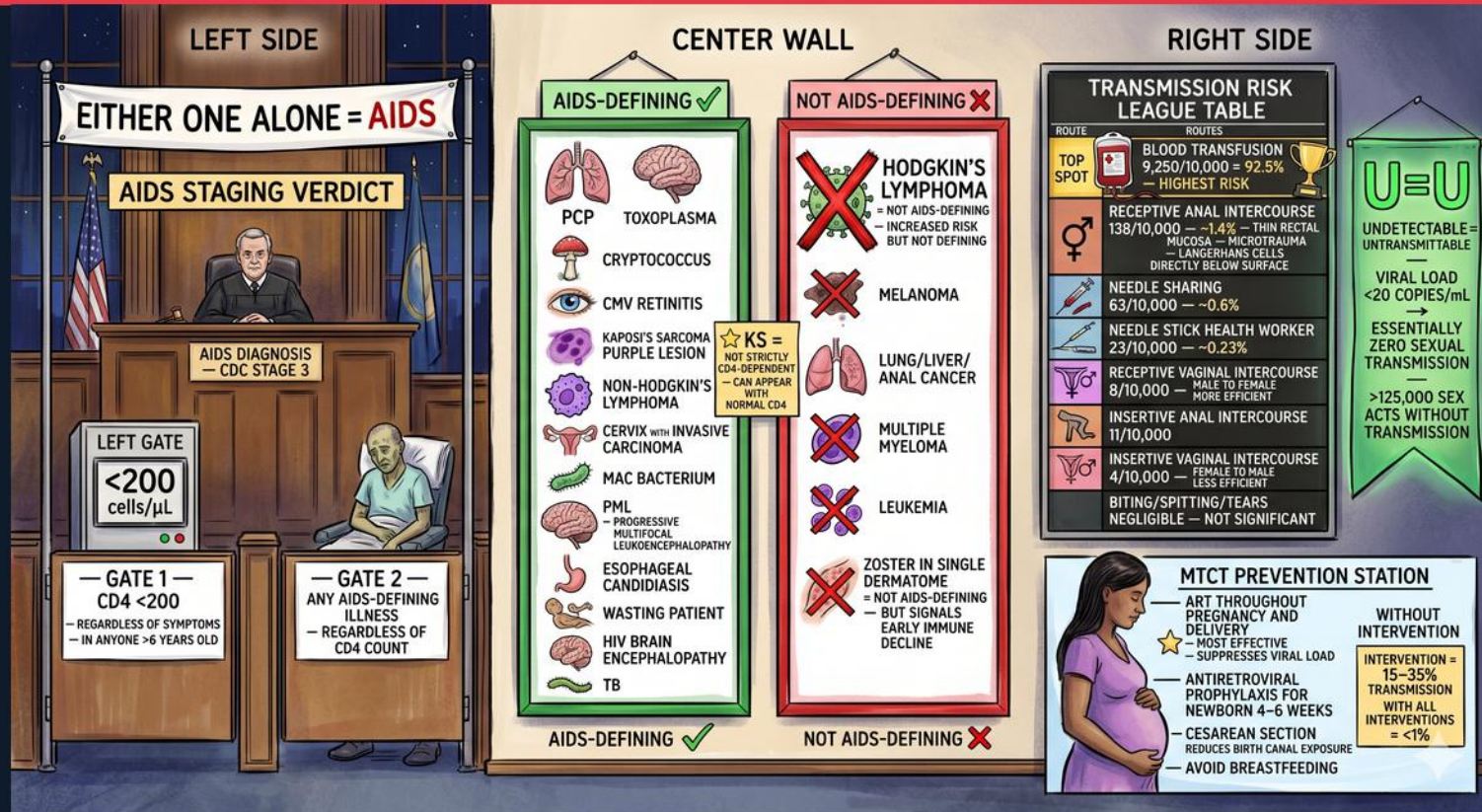


HIV — AIDS-Defining Conditions & Staging Verdict





Judge: staging verdict

WHAT YOU SEE

Judge banner: EITHER ONE ALONE = AIDS, CDC Stage 3

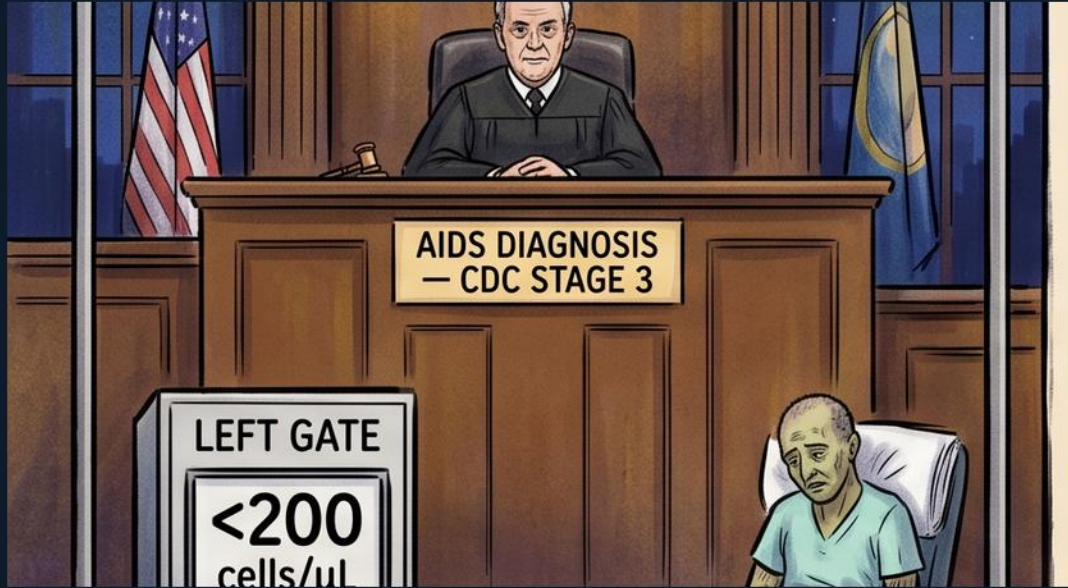
WHAT IT ENCODES

AIDS = CDC Stage 3, diagnosed when EITHER Gate 1 (CD4 <200 cells/ μ L or CD4% <14) OR Gate 2 (a documented AIDS-defining opportunistic illness) is met — either one alone is sufficient in a person with confirmed HIV ≥ 6 years old. Once a patient meets a Stage 3 criterion they remain classified as AIDS even if CD4 later rises on ART (the staging is not reversed).

THE BOARD TRAP

Wrong answer: 'a patient needs BOTH CD4 <200 AND an AIDS-defining illness to be diagnosed with AIDS.'

Discriminator: the two gates are independent; meeting EITHER one alone establishes AIDS. Requiring both is the classic distractor.



Gate 1: CD4 <200

WHAT YOU SEE

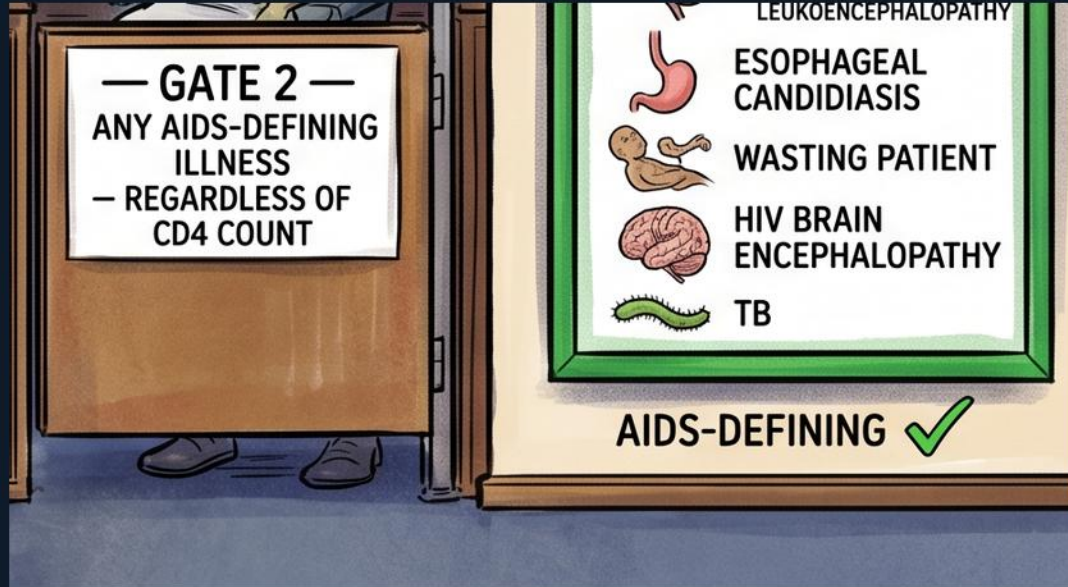
Left gate <200 cells/ μ L, CD4 <200 regardless of symptoms at any age >6yr

WHAT IT ENCODES

A CD4 count <200 cells/ μ L (or CD4 percentage <14%) defines AIDS regardless of how well the patient feels. This threshold also triggers Pneumocystis (PCP) prophylaxis with TMP-SMX. The CD4 count gauges current immunodeficiency and dictates which opportunistic infections are likely and which prophylaxis to start.

THE BOARD TRAP

Wrong answer: 'an asymptomatic patient with CD4 150 is just HIV-positive, not AIDS.' Discriminator: CD4 <200 is an AIDS-defining criterion on its own — symptoms are irrelevant. This patient has AIDS (CDC Stage 3) and needs PCP prophylaxis plus ART.



Gate 2: AIDS illness

WHAT YOU SEE

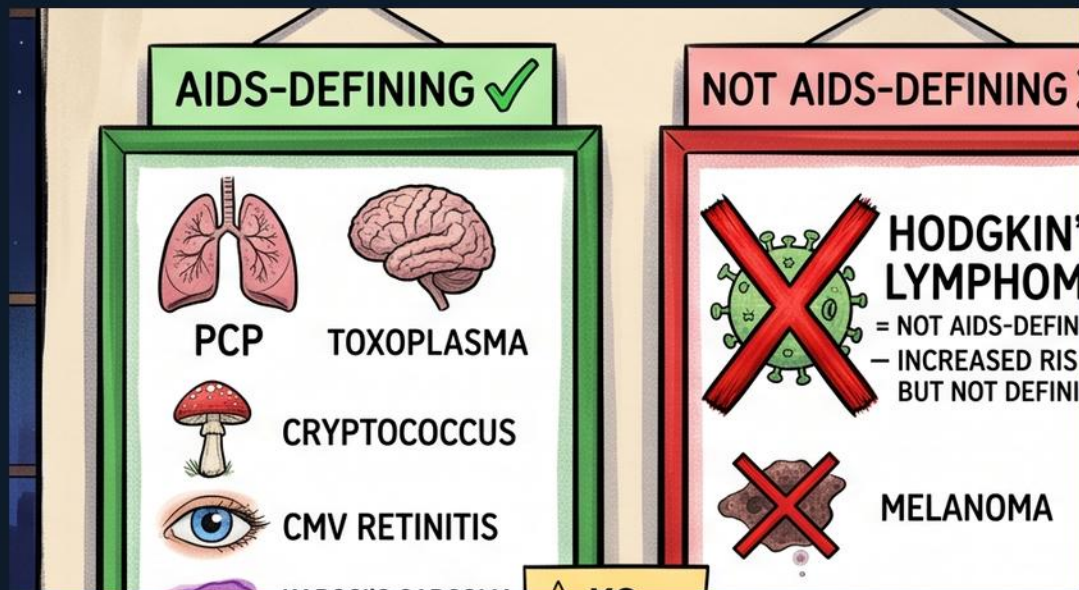
Gate 2: any AIDS-defining illness regardless of CD4 count

WHAT IT ENCODES

Any qualifying AIDS-defining opportunistic illness establishes AIDS even if the CD4 count is normal/high. Some illnesses (e.g., esophageal candidiasis, PCP, Kaposi sarcoma) can occasionally occur at higher CD4 counts and still count. The diagnosis is clinical/illness-based, independent of the number.

THE BOARD TRAP

Wrong answer: 'this patient has PCP but a CD4 of 320, so it can't be AIDS.' Discriminator: an AIDS-defining illness diagnoses AIDS regardless of CD4 count. A normal CD4 does NOT exclude AIDS if a qualifying OI is present.



PCP / PJP

WHAT YOU SEE

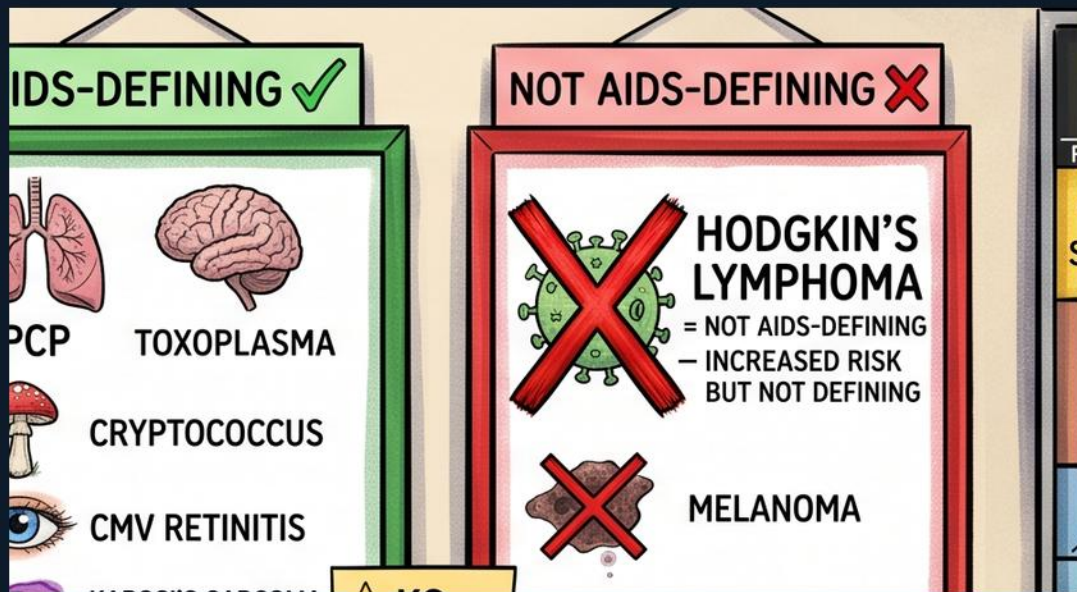
Lungs labeled PCP under AIDS-defining green column

WHAT IT ENCODES

Pneumocystis jirovecii pneumonia (PCP/PJP): the prototypical AIDS-defining illness at CD4 <200. Presents with subacute dyspnea, dry cough, hypoxia with bilateral interstitial/ground-glass infiltrates and elevated LDH. Treat with TMP-SMX (21 days); ADD adjunctive corticosteroids if PaO₂ <70 mmHg or A-a gradient ≥35. Prophylaxis (TMP-SMX) starts at CD4 <200.

THE BOARD TRAP

Wrong answer: 'start TMP-SMX alone for severe PCP with PaO₂ 60.' Discriminator: in hypoxemic PCP (PaO₂ <70 or A-a gradient ≥35) you MUST add corticosteroids before/with antibiotics to prevent treatment-related respiratory worsening — omitting steroids is the trap.



Toxoplasma

WHAT YOU SEE

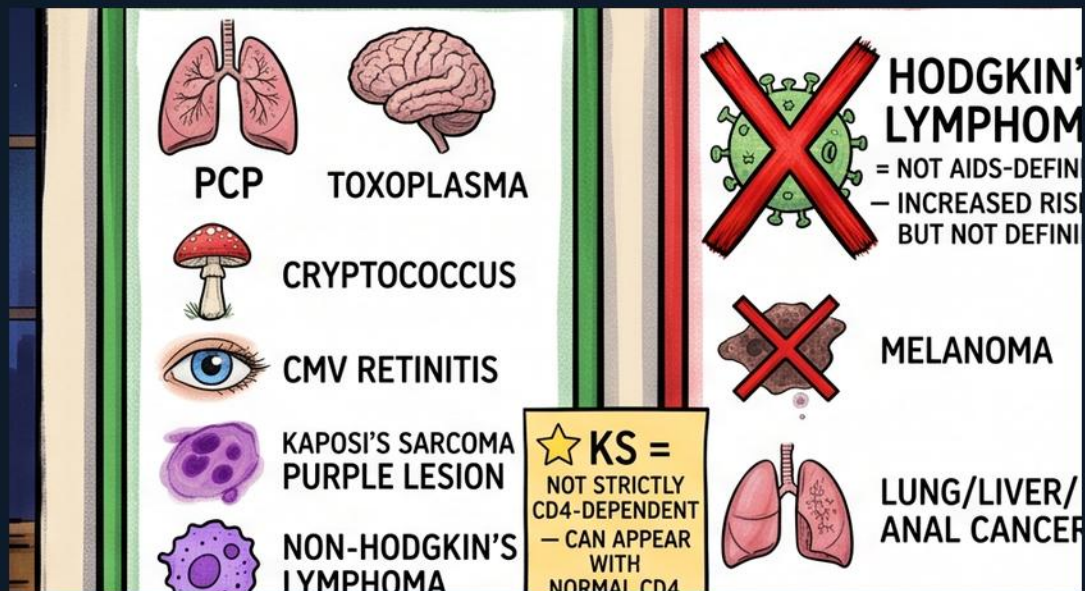
Brain icon labeled Toxoplasma

WHAT IT ENCODES

CNS toxoplasmosis (CD4 <100): MULTIPLE ring-enhancing lesions on contrast MRI (basal ganglia/gray-white junction), positive Toxoplasma IgG, mass effect. Treat empirically with sulfadiazine + pyrimethamine + leucovorin; clinical/radiographic improvement in ~2 weeks confirms it. TMP-SMX prophylaxis (the PCP regimen) also covers toxo at CD4 <100.

THE BOARD TRAP

Wrong answer: 'a single ring-enhancing brain lesion with positive Toxo IgG must be toxoplasmosis.' Discriminator: toxo classically gives MULTIPLE lesions; a SOLITARY lesion (especially if it fails empiric therapy or EBV PCR is positive in CSF) favors primary CNS lymphoma — biopsy if no response to anti-toxo treatment.



Cryptococcus

WHAT YOU SEE

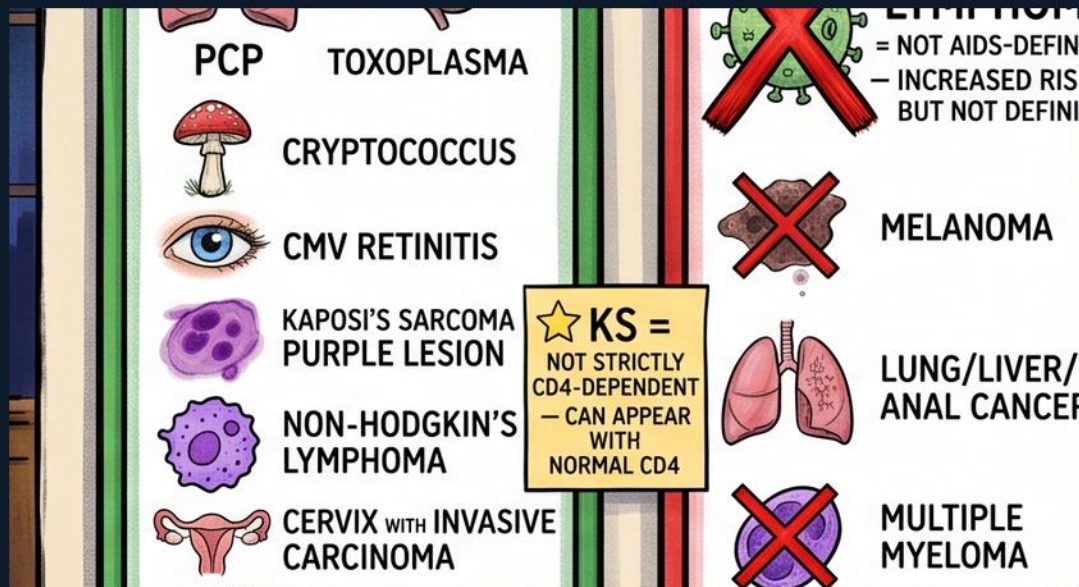
Cryptococcus listed under AIDS-defining

WHAT IT ENCODES

Cryptococcal disease, especially extrapulmonary cryptococcal meningitis (CD4 <100, often <50): subacute headache, fever, raised intracranial pressure. Diagnose with serum/CSF cryptococcal antigen (CrAg) and India ink. Treat with induction liposomal amphotericin B + flucytosine, then fluconazole. MANAGE ICP with serial therapeutic LPs.

THE BOARD TRAP

Wrong answer: 'treat cryptococcal meningitis with high-pressure headaches using a shunt/acetazolamide first.' Discriminator: elevated ICP in crypto meningitis is managed with repeated therapeutic lumbar punctures (drain CSF), not acetazolamide/mannitol; failure to control ICP is a leading cause of death.



CMV retinitis

WHAT YOU SEE

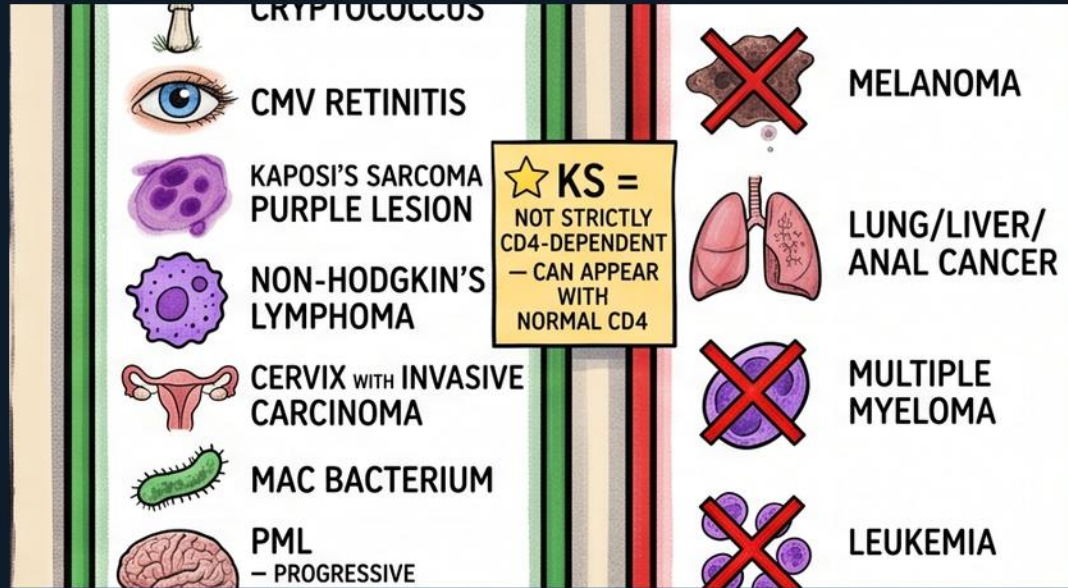
Eye icon CMV retinitis

WHAT IT ENCODES

CMV end-organ disease (CD4 <50): retinitis is most common — painless vision loss/floaters with 'pizza-pie' hemorrhages and fluffy yellow-white perivascular exudates on fundoscopy. Also causes colitis and esophagitis. Treat with ganciclovir/valganciclovir (or foscarnet). ART-driven immune recovery can trigger immune-recovery uveitis (an IRIS phenomenon).

THE BOARD TRAP

Wrong answer: 'an HIV patient with floaters and CD4 90 most likely has CMV retinitis.' Discriminator: CMV retinitis classically occurs at CD4 <50, not 90. Also note painless visual change with pizza-pie hemorrhages — painful red eye points elsewhere (e.g., VZV/HSV retinitis or other uveitis).



Kaposi sarcoma

WHAT YOU SEE

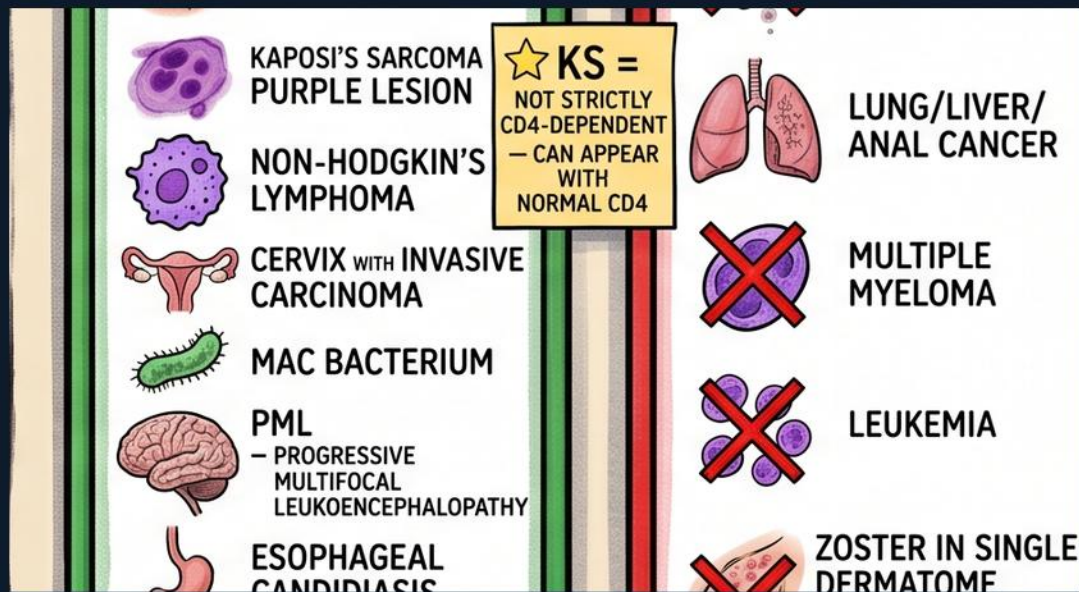
Purple skin lesion, Kaposi's sarcoma

WHAT IT ENCODES

Kaposi sarcoma is caused by HHV-8 (human herpesvirus 8) and is AIDS-defining: violaceous/purple non-blanching plaques and nodules on skin, oral mucosa (palate), GI tract, and lungs. Histology shows spindle cells and slit-like vascular spaces. Cornerstone treatment is ART (immune restoration); add systemic chemotherapy (liposomal doxorubicin) for visceral/extensive disease.

THE BOARD TRAP

Wrong answer: 'Kaposi sarcoma is caused by HHV-6 / EBV.'
Discriminator: Kaposi is HHV-8 (KSHV). EBV is linked to the lymphomas and oral hairy leukoplakia; HHV-6 is roseola. Mixing up the herpesvirus is a frequent distractor.



Lymphomas (NHL/CNS)

WHAT YOU SEE

Non-Hodgkin & primary CNS lymphoma listed AIDS-defining

WHAT IT ENCODES

AIDS-defining lymphomas are aggressive B-cell non-Hodgkin lymphomas: Burkitt, diffuse large B-cell/immunoblastic, and primary CNS lymphoma (nearly 100% EBV-driven, typically CD4 <50, often a SOLITARY weakly/homogeneously enhancing periventricular lesion). EBV PCR in CSF and thallium-SPECT/PET help distinguish CNS lymphoma from toxoplasmosis.

THE BOARD TRAP

Wrong answer: 'Hodgkin lymphoma is AIDS-defining because HIV increases its risk.' Discriminator: although HIV raises Hodgkin incidence, Hodgkin lymphoma is NOT AIDS-defining. Only certain aggressive B-cell NHLs (Burkitt, DLBCL/immunoblastic, primary CNS lymphoma) qualify.

	NON-HODGKIN'S LYMPHOMA	— CAN APPEAR WITH NORMAL CD4		ANAL CANCER
	CERVIX WITH INVASIVE CARCINOMA			MULTIPLE MYELOMA
	MAC BACTERIUM			LEUKEMIA
	PML — PROGRESSIVE MULTIFOCAL LEUKOENCEPHALOPATHY			ZOSTER IN SINGLE DERMATOME = NOT AIDS-DEFINING — BUT SIGNALS EARLY IMMUNE DECLINE
	ESOPHAGEAL CANDIDIASIS			
	WASTING PATIENT			
	HIV BRAIN			

Invasive cervical CA

WHAT YOU SEE

Cervix → invasive carcinoma

WHAT IT ENCODES

Invasive cervical carcinoma (HPV-driven) is AIDS-defining. HIV-positive women have accelerated HPV-related disease and need more frequent cervical cancer screening. Pre-invasive lesions (dysplasia/CIN) are common but do NOT meet the AIDS-defining bar — only INVASIVE cancer counts.

THE BOARD TRAP

Wrong answer: 'high-grade cervical dysplasia (CIN 2/3) in an HIV patient is AIDS-defining.' Discriminator: only INVASIVE cervical carcinoma is AIDS-defining; cervical intraepithelial neoplasia/dysplasia is not. The same 'invasive vs in-situ' logic distinguishes the answer.

	CERVIX WITH INVASIVE CARCINOMA		MULTIPLE MYELOMA
	MAC BACTERIUM		LEUKEMIA
	PML — PROGRESSIVE MULTIFOCAL LEUKOENCEPHALOPATHY		ZOSTER IN SINGLE DERMATOME = NOT AIDS-DEFINING — BUT SIGNALS EARLY IMMUNE DECLINE
	ESOPHAGEAL CANDIDIASIS		
	WASTING PATIENT		
	HIV BRAIN ENCEPHALOPATHY		
	TB		

MAC

WHAT YOU SEE

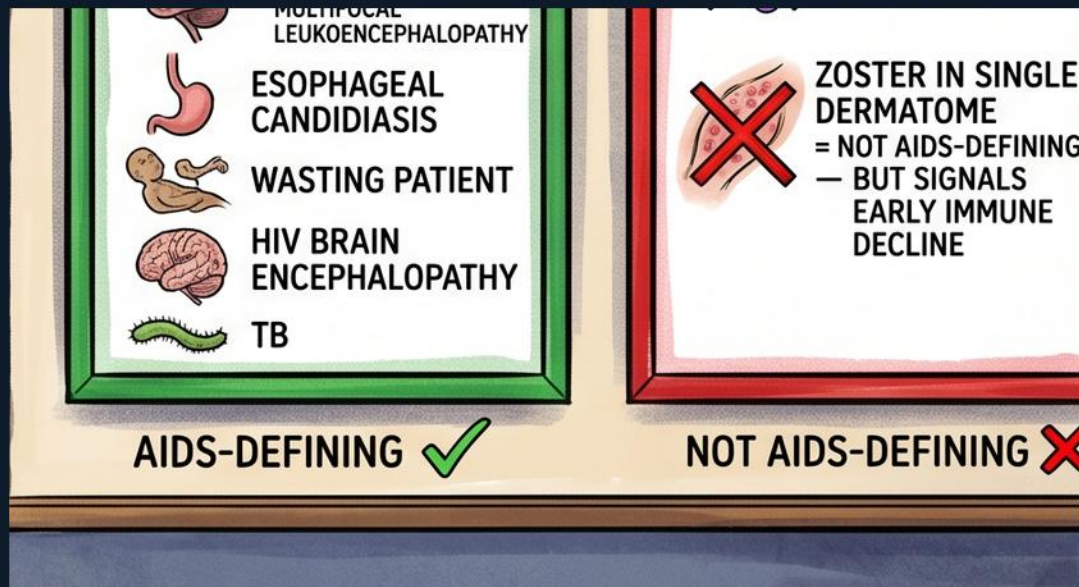
MAC bacterium listed

WHAT IT ENCODES

Disseminated Mycobacterium avium complex (CD4 <50): fevers, night sweats, weight loss, diarrhea, hepatosplenomegaly, cytopenias, markedly elevated alkaline phosphatase; diagnosed by blood/marrow culture. Treat with a macrolide (azithromycin/clarithromycin) + ethambutol ± rifabutin. Current guidance: routine primary MAC prophylaxis is largely deferred if ART is started promptly.

THE BOARD TRAP

Wrong answer: 'every patient with CD4 <50 needs azithromycin MAC prophylaxis.' Discriminator: with modern early ART, routine primary MAC prophylaxis is no longer recommended for those starting effective ART; the older 'prophylaxis at CD4 <50' teaching has been revised. ART is the priority.



Esophageal candidiasis

WHAT YOU SEE

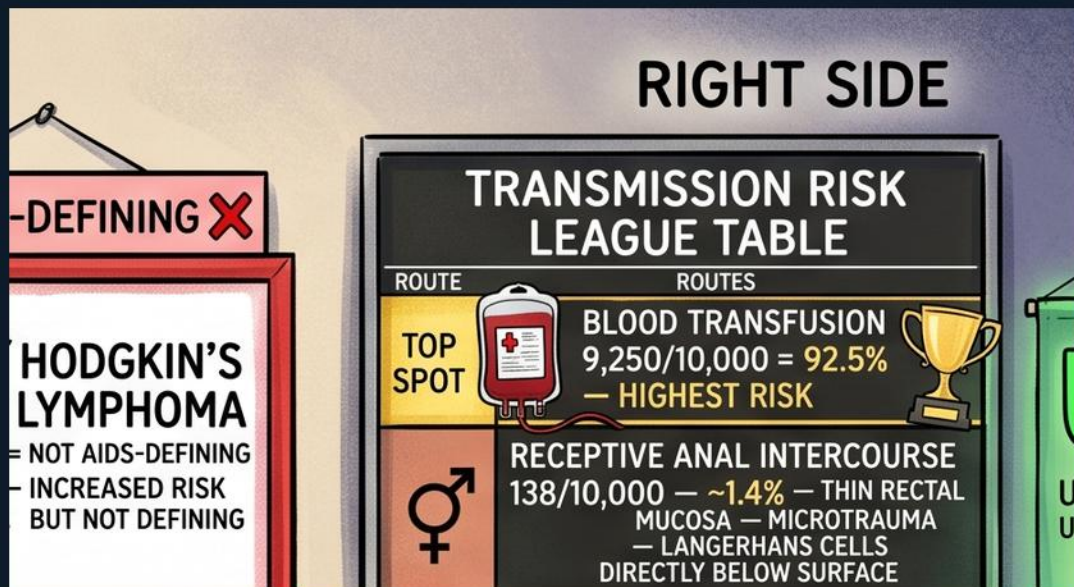
Esophageal candidiasis listed AIDS-defining

WHAT IT ENCODES

Esophageal (or tracheal/bronchial/pulmonary) candidiasis is AIDS-defining and typically presents with odynophagia/dysphagia at CD4 <200. Diagnosis is usually clinical (HIV + esophageal symptoms + oral thrush) with empiric systemic fluconazole; endoscopy is reserved for non-responders to exclude CMV/HSV esophagitis.

THE BOARD TRAP

Wrong answer: 'oral thrush (oropharyngeal candidiasis) is AIDS-defining.' Discriminator: oral candidiasis alone is NOT AIDS-defining — it must extend to the ESOPHAGUS or lower respiratory tract (trachea/bronchi/lungs) to qualify. Oral thrush is, however, a marker that CD4 is likely <200.



NOT AIDS-defining

WHAT YOU SEE

Red X column: Hodgkin lymphoma, melanoma, lung/liver/anal CA, myeloma, leukemia

WHAT IT ENCODES

HIV increases the incidence of many non-AIDS malignancies that are NOT AIDS-defining: Hodgkin lymphoma, lung/liver/anal cancer, melanoma, multiple myeloma, and leukemia. These reflect chronic immune activation and shared risk factors (smoking, HPV, viral hepatitis) but do not meet CDC Stage 3 criteria.

THE BOARD TRAP

Wrong answer: 'multiple myeloma / anal cancer in an HIV patient = AIDS-defining.' Discriminator: these are increased-incidence but NON-AIDS-defining cancers. The frequent exam trap pair is Hodgkin lymphoma and multiple myeloma — both are NOT AIDS-defining despite higher incidence.

MULTIPLE
MYELOMA

LEUKEMIA

ZOSTER IN SINGLE
DERMATOME
NOT AIDS-DEFINING
BUT SIGNALS
EARLY IMMUNE
DECLINE

	11/10,000
	INSERTIVE VAGINAL INTERCOURSE 4/10,000 — FEMALE TO MALE LESS EFFICIENT
	BITING/SPITTING/TEARS NEGLECTIBLE — NOT SIGNIFICANT

MTCT PREVENTION STR



- ART THROUGHOUT PREGNANCY AND DELIVERY
 - ★ — MOST EFFECTIVE
 - SUPPRESSES VIRAL LOAD
- ANTIRETROVIRAL

Localized zoster

WHAT YOU SEE

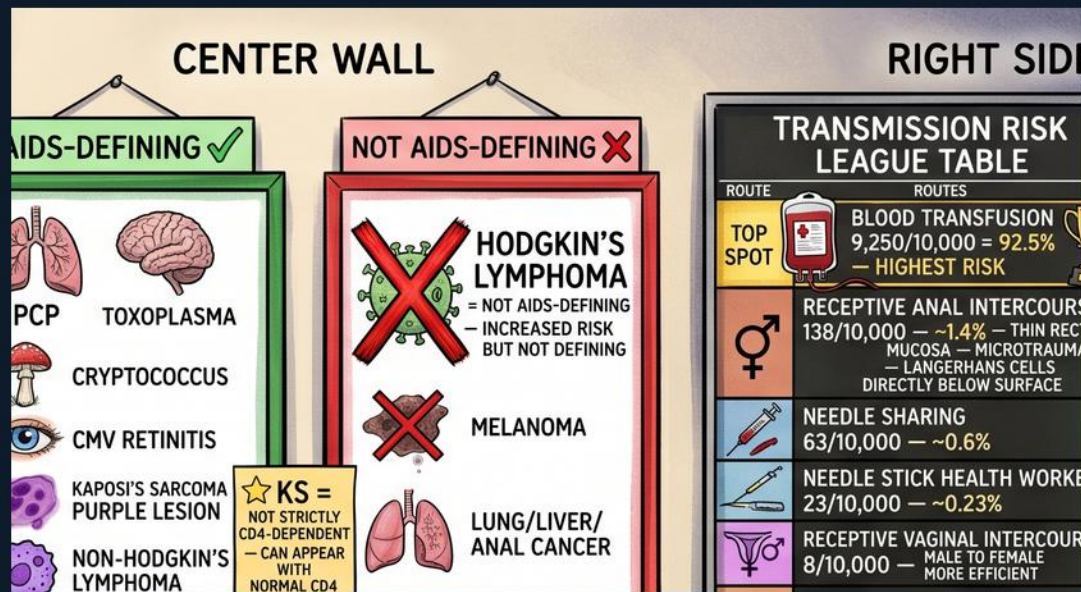
Zoster in single dermatome marked NOT AIDS-defining

WHAT IT ENCODES

Single-dermatome herpes zoster, generalized wasting without an opportunistic cause, and general decline are not by themselves AIDS-defining. However, multidermatomal/disseminated zoster or recurrent episodes signal advancing immunodeficiency, and HIV-related WASTING syndrome (involuntary >10% weight loss plus chronic diarrhea/fever) IS a defined AIDS condition.

THE BOARD TRAP

Wrong answer: 'any herpes zoster in an HIV patient is AIDS-defining.' Discriminator: localized single-dermatome zoster is NOT AIDS-defining; only disseminated/multidermatomal involvement is concerning, and zoster generally is not on the core AIDS-defining illness list the way wasting syndrome (with weight loss) is.



OI prophylaxis thresholds

WHAT YOU SEE

'OI Prophylaxis Threshold Tower' — CD4 200 / 100 / 50 floors

WHAT IT ENCODES

CD4-based prophylaxis ladder: at CD4 <200 start TMP-SMX for PCP; at CD4 <100 (with positive Toxo IgG) TMP-SMX also covers toxoplasmosis (one pill covers both); CMV (<50) and MAC (<50) prophylaxis are now generally deferred in favor of prompt ART. Prophylaxis can be discontinued once ART raises CD4 above the threshold for a sustained period (e.g., PCP prophylaxis stop at CD4 >200 for 3 months).

THE BOARD TRAP

Wrong answer: 'continue PCP prophylaxis indefinitely regardless of CD4 recovery.' Discriminator: primary PCP prophylaxis is stopped once ART sustains CD4 >200 for ≥3 months — lifelong prophylaxis is unnecessary and the threshold-based stop is the tested point.

R WALL

NOT AIDS-DEFINING ❌

HODGKIN'S LYMPHOMA
= NOT AIDS-DEFINING
— INCREASED RISK
BUT NOT DEFINING

MELANOMA

S =
STRICTLY
PENDENT
APPEAR
TH
AL CD4

**LUNG/LIVER/
ANAL CANCER**

RIGHT SIDE

TRANSMISSION RISK LEAGUE TABLE	
ROUTE	ROUTES
TOP SPOT	BLOOD TRANSFUSION 9,250/10,000 = 92.5% — HIGHEST RISK
♀	RECEPTIVE ANAL INTERCOURSE 138/10,000 — ~1.4% — THIN RECTAL MUCOSA — MICROTRAUMA — LANGERHANS CELLS DIRECTLY BELOW SURFACE
📌	NEEDLE SHARING 63/10,000 — ~0.6%
📌	NEEDLE STICK HEALTH WORKER 23/10,000 — ~0.23%
♀	RECEPTIVE VAGINAL INTERCOURSE 8/10,000 — MALE TO FEMALE MORE EFFICIENT

U=U
UNDETECTABLE =
UNTRANSMITTABLE
—
VIRAL LOAD
<20 COPIES/mL
→
ESSENTIALLY
ZERO SEXUAL
TRANSMISSION
—
>125,000 SEX

Start ART in everyone

WHAT YOU SEE

'ART Timing Station' green light — start ART in everyone regardless of CD4, ASAP

WHAT IT ENCODES

Current standard: start ART in ALL people with HIV as soon as possible after diagnosis, regardless of CD4 count or viral load — this replaced the old 'wait for CD4 <350/<500' teaching. Early ART improves survival, reduces non-AIDS complications, and achieves U=U (undetectable = untransmittable: a sustained undetectable viral load means HIV is not sexually transmitted).

THE BOARD TRAP

Wrong answer: 'defer ART until CD4 falls below 350/500.'
Discriminator: that is OUTDATED. Modern guidelines start ART for everyone with HIV immediately at diagnosis irrespective of CD4. The only nuance is timing in certain CNS OIs (e.g., cryptococcal meningitis), where ART may be briefly delayed to avoid IRIS.

Summary — every symbol

1. Judge: staging verdict

AIDS = CDC Stage 3, diagnosed when EITHER Gate 1 (CD4 <200 cells/μL or CD4% <14) OR Gate 2 (a documented AIDS-defining opportunistic illness) is met — either one alone is sufficient in a person with confirmed HIV ≥6 years old

3. Gate 2: AIDS illness

Any qualifying AIDS-defining opportunistic illness establishes AIDS even if the CD4 count is normal/high

5. Toxoplasma

CNS toxoplasmosis (CD4 <100): MULTIPLE ring-enhancing lesions on contrast MRI (basal ganglia/gray-white junction), positive Toxoplasma IgG, mass effect

7. CMV retinitis

CMV end-organ disease (CD4 <50): retinitis is most common — painless vision loss/floaters with 'pizza-pie' hemorrhages and fluffy yellow-white perivascular exudates on fundoscopy

9. Lymphomas (NHL/CNS)

AIDS-defining lymphomas are aggressive B-cell non-Hodgkin lymphomas: Burkitt, diffuse large B-cell/immunoblastic, and primary CNS lymphoma (nearly 100% EBV-driven, typically CD4 <50, often a SOLITARY weakly/homogeneously enhancing periventricular lesion)

11. MAC

Disseminated Mycobacterium avium complex (CD4 <50): fevers, night sweats, weight loss, diarrhea, hepatosplenomegaly, cytopenias, markedly elevated alkaline phosphatase; diagnosed by blood/marrow culture

13. NOT AIDS-defining

HIV increases the incidence of many non-AIDS malignancies that are NOT AIDS-defining: Hodgkin lymphoma, lung/liver/anal cancer, melanoma, multiple myeloma, and leukemia

15. OI prophylaxis thresholds

CD4-based prophylaxis ladder: at CD4 <200 start TMP-SMX for PCP; at CD4 <100 (with positive Toxo IgG) TMP-SMX also covers toxoplasmosis (one pill covers both); CMV (<50) and MAC (<50) prophylaxis are now generally deferred in favor of prompt ART

2. Gate 1: CD4 <200

A CD4 count <200 cells/μL (or CD4 percentage <14%) defines AIDS regardless of how well the patient feels

4. PCP / PJP

Pneumocystis jirovecii pneumonia (PCP/PJP): the prototypical AIDS-defining illness at CD4 <200

6. Cryptococcus

Cryptococcal disease, especially extrapulmonary cryptococcal meningitis (CD4 <100, often <50): subacute headache, fever, raised intracranial pressure

8. Kaposi sarcoma

Kaposi sarcoma is caused by HHV-8 (human herpesvirus 8) and is AIDS-defining: violaceous/purple non-blanching plaques and nodules on skin, oral mucosa (palate), GI tract, and lungs

10. Invasive cervical CA

Invasive cervical carcinoma (HPV-driven) is AIDS-defining

12. Esophageal candidiasis

Esophageal (or tracheal/bronchial/pulmonary) candidiasis is AIDS-defining and typically presents with odynophagia/dysphagia at CD4 <200

14. Localized zoster

Single-dermatome herpes zoster, generalized wasting without an opportunistic cause, and general decline are not by themselves AIDS-defining

16. Start ART in everyone

Current standard: start ART in ALL people with HIV as soon as possible after diagnosis, regardless of CD4 count or viral load — this replaced the old 'wait for CD4 <350/<500' teaching